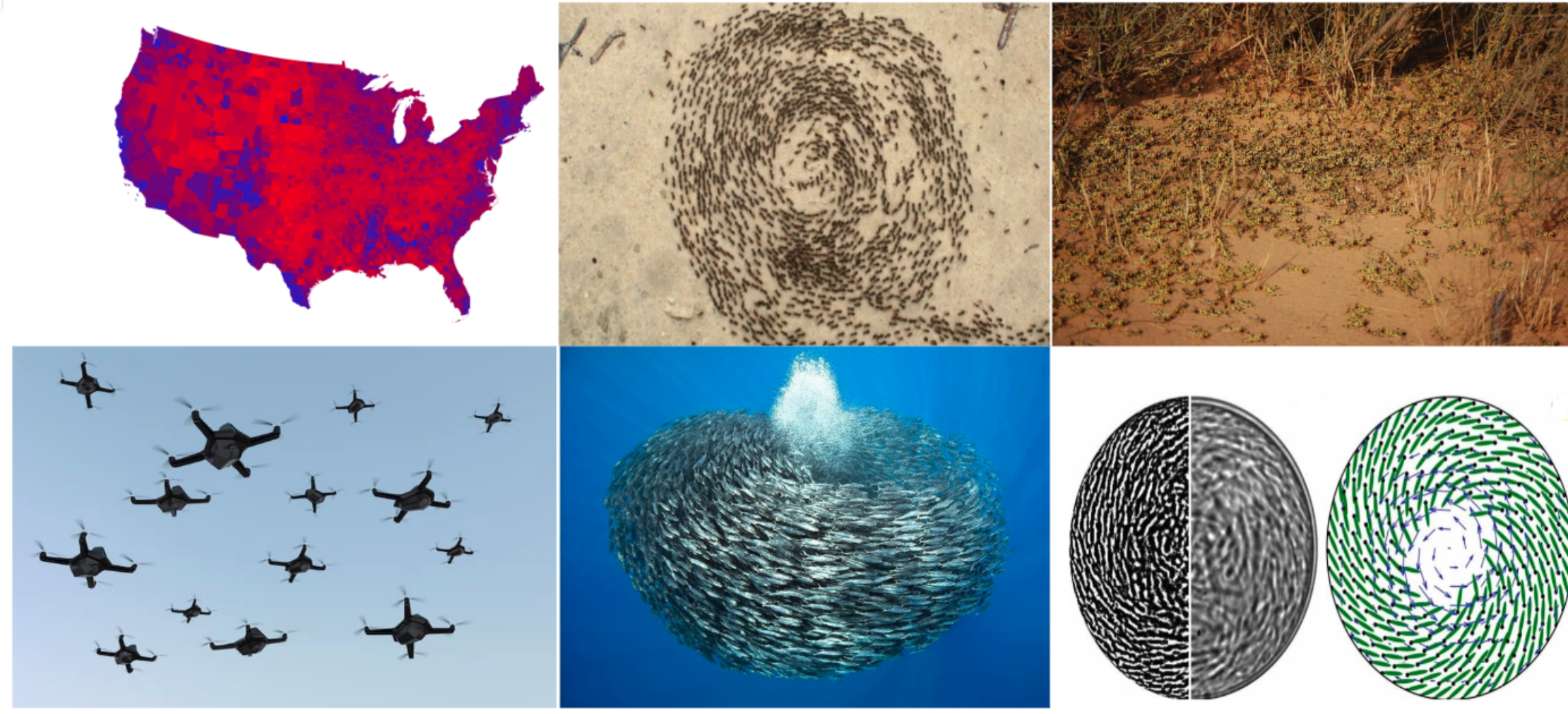


Introduction

Collective Systems in Nature

Collective behavior encompasses the study of how individuals within a group exhibit behaviors distinct from their actions in isolation. This field explores emergent patterns, spontaneous actions, and the influence of group dynamics on behavior. Understanding collective behavior sheds light on phenomena such as biological motion, rioting, and social behavior, offering insights into the complexities of group dynamics and societal interactions.

The dynamics between agents can manifest via simple mechanisms, characterized by basic forces of attraction or repulsion, or in greater intricacy, arising from combinations of these fundamental interactions.



Objective

In this work, we will extend the algorithm in [3] to frameworks that contain both collective and self-propulsion terms since some collective system dynamics arise as a combination of both collective influence and self-propulsion.

These systems are defined as groups of individual entities capable of self-propulsion, which interact with each other, leading to coordinated, emergent behavior.

We update the general framework with a new term ψ to embed the self-propulsion features.

$$\dot{x}_i = \psi(x_i) + \frac{1}{N} \sum_{j=1}^N \phi(\|x_j - x_i\|)(x_j - x_i) \quad (1)$$

Future Work

- We plan to extend the algorithm to incorporate a non-parametric self-propulsion term in future work.
- We will investigate methods for selecting the best-fit model from the set of all possible variations based on the available data.

$$\dot{x}_i = \psi(x_i) + \frac{1}{N} \sum_{j=1}^N \phi_E(\|x_j - x_i\|)(x_j - x_i) \quad (2)$$

$$\ddot{x}_i = \psi(x_i, \dot{x}_i) + \frac{1}{N} \left[\sum_{j=1}^N \phi_E(\|x_j - x_i\|)(x_j - x_i) + \sum_{j=1}^N \phi_A(\|x_j - x_i\|)(\dot{x}_j - \dot{x}_i) \right] \quad (3)$$

Given the data, we get the system approximations for frameworks with combinations of ψ , ϕ_E and ϕ_A and for 1st order and 2nd order frameworks. To find the optimal model we will be using Knee/Elbow Estimation Based on First Derivative Threshold. [1]

Implementation

The goal of the present poster is to extend the algorithm when the self-propulsion term is parametric. In future work, we plan to address the case when the self-propulsion term is non-parametric.

The objective is to estimate the interaction kernel ϕ of (1) by selecting a candidate function $\varphi \in \mathcal{H}$ from a suitable hypothesis space $\mathcal{H}$ of functions on $\mathbb{R}_+$ with $\dim(\mathcal{H}) = n$ and to estimate the parameters of the self-propulsion term Θ from some compact subset $\mathcal{K}$.

$$\dot{x}_i^{(m)} = \psi(x_i^{(m)}, \Theta) + \frac{1}{N} \sum_{j=1}^N \phi(\|x_j^{(m)} - x_i^{(m)}\|)(x_j^{(m)} - x_i^{(m)}) \quad (4)$$

$$:= \psi(x_i^{(m)}, \Theta) + f_\phi(x_i^{(m)}) \quad (5)$$

Consider the empirical error functional for any φ and θ

$$\varepsilon_{L,M}(\varphi, \theta) := \frac{1}{LMN} \sum_{l,m,i=1}^{L,M,N} \|\dot{x}_i^{(m)}(t_l) - \psi(x_i^{(m)}(t_l), \theta) - f_\varphi(x_i^{(m)}(t_l))\|_2^2 \quad (6)$$

The estimators are defined as a minimizer of $\varepsilon_{L,M}$ over $\varphi \in \mathcal{H}$, and $\theta \in \mathcal{K} \subseteq \mathbb{R}^k$ for some compact subset $\mathcal{K}$

$$\hat{\phi}, \hat{\Theta} := \arg \min_{\varphi \in \mathcal{H}, \theta \in \mathcal{K}} \varepsilon_{L,M}(\varphi, \theta)$$

Simulations

1st order system - Kuramoto

A well-known model of synchronization that can also describe the behavior of a large collection of coupled oscillators:

$$\dot{\theta}_i = \omega_i + \frac{1}{N} \sum_{j=1}^N K_{ij} \sin(\theta_j - \theta_i)$$

where:

N : number of oscillators

K_{ij} : coupling constant

θ_i : phase

ω_i : intrinsic frequency

2nd order system- self-propelled particle (SPP) system

This system describes the swarming of multi-agents with self-propulsion:

$$\begin{aligned} \dot{x}_i &= v_i \\ \dot{v}_i &= [(\alpha - \beta|v_i|^2)v_i - \nabla_i U(x_i)] \end{aligned}$$

with the generalized Morse potential given as

$$U(x_i) = \sum_{j \neq i}^N C_r e^{-|x_i - x_j|/l_r} - C_a e^{-|x_i - x_j|/l_a}$$

where:

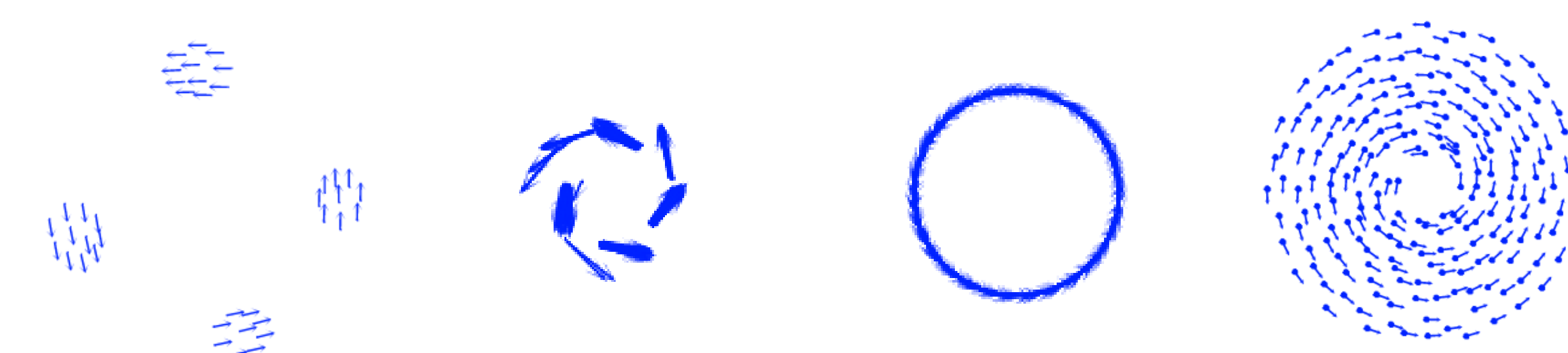
α : self-propulsion strength

l_a, l_r : attractive and repulsive potential ranges respectively

β : nonlinear damping

C_a, C_r : attractive and repulsive potential amplitudes respectively

The model performs different behaviors for different parameters[2].

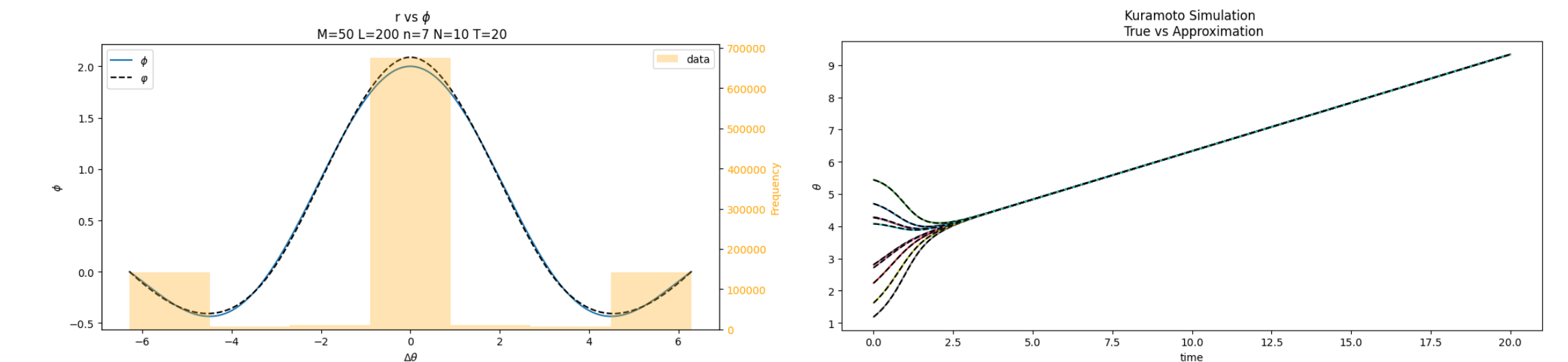


Results

1st order system - Kuramoto

$$\dot{\theta}_i = \underbrace{\omega_i}_{\psi} + \frac{1}{N} \sum_{j=1}^N K \underbrace{\frac{\sin(\theta_j - \theta_i)}{(\theta_j - \theta_i)}}_{\phi} (\theta_j - \theta_i)$$

M	L	t ₀	T	N	n	d	ω
50	200	0	20	10	7	3	0.3



$$\hat{\Theta} = 0.3000036$$

Here, $\psi = \Theta, \psi = 0.3$

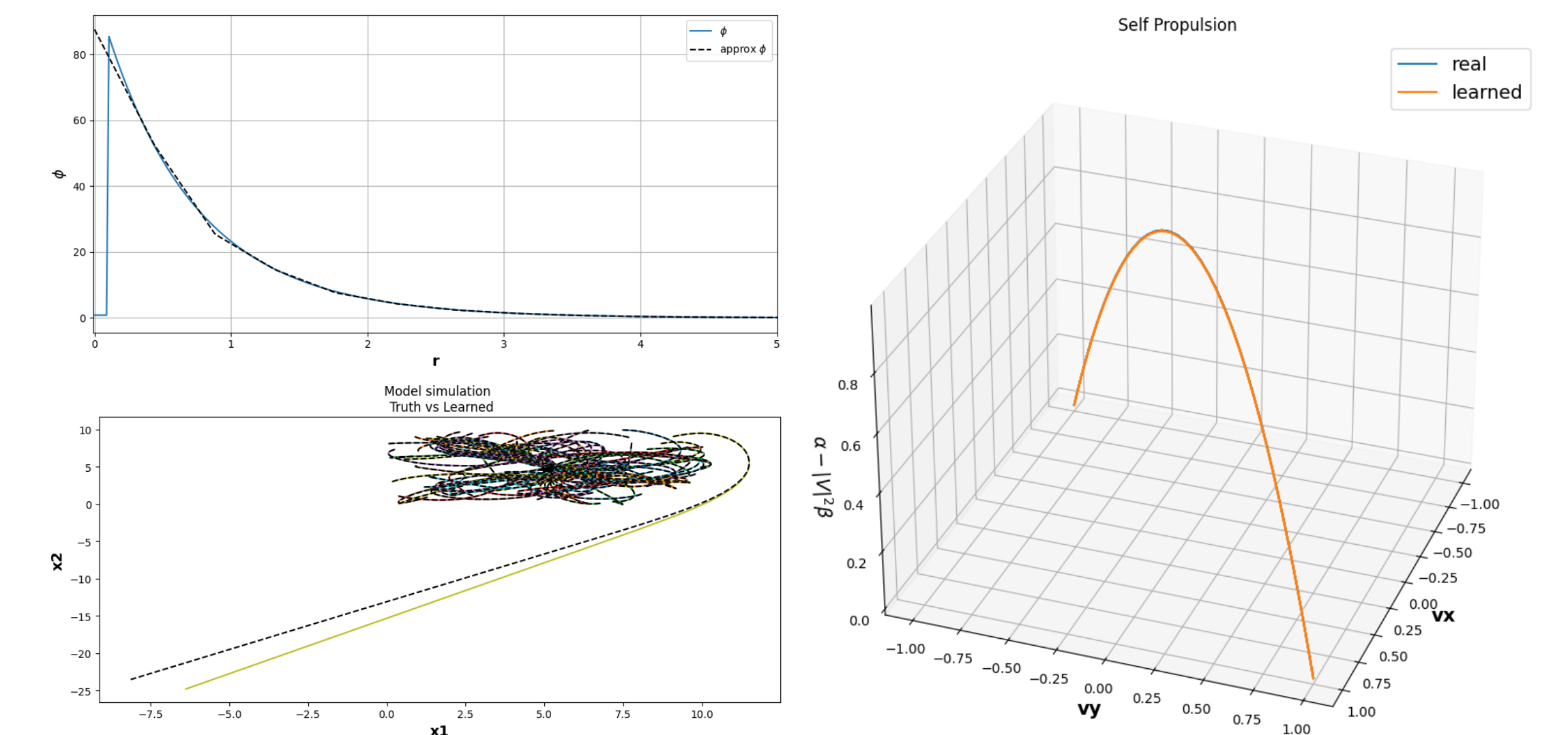
2nd order system- self-propelled particle (SPP) system

Let $r_{ij} = x_i - x_j$, the system becomes:

M	L	t ₀	T	N	n	d
20	90	0	30	60	100	5

$$\dot{x}_i = v_i$$

$$\dot{v}_i = \underbrace{(\alpha - \beta|v_i|^2)v_i}_{\psi} + \sum_{j=1, j \neq i}^N \underbrace{\left(\frac{C_r e^{-|r_{ij}|/l_r}}{l_r} - \frac{C_a e^{-|r_{ij}|/l_a}}{l_a} \right)}_{\phi} \frac{r_{ij}}{|r_{ij}|}$$



References

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- [2] M. R. D’Orsogna et al. “Self-Propelled Particles with Soft-Core Interactions: Patterns, Stability, and Collapse”. In: *Phys. Rev. Lett.* 96 (10 Mar. 2006), p. 104302. doi: 10.1103/PhysRevLett.96.104302. url: <https://link.aps.org/doi/10.1103/PhysRevLett.96.104302>.
- [3] Fei Lu et al. “Nonparametric inference of interaction laws in systems of agents from trajectory data”. In: *Proceedings of the National Academy of Sciences* 116.29 (2019), pp. 14424–14433.